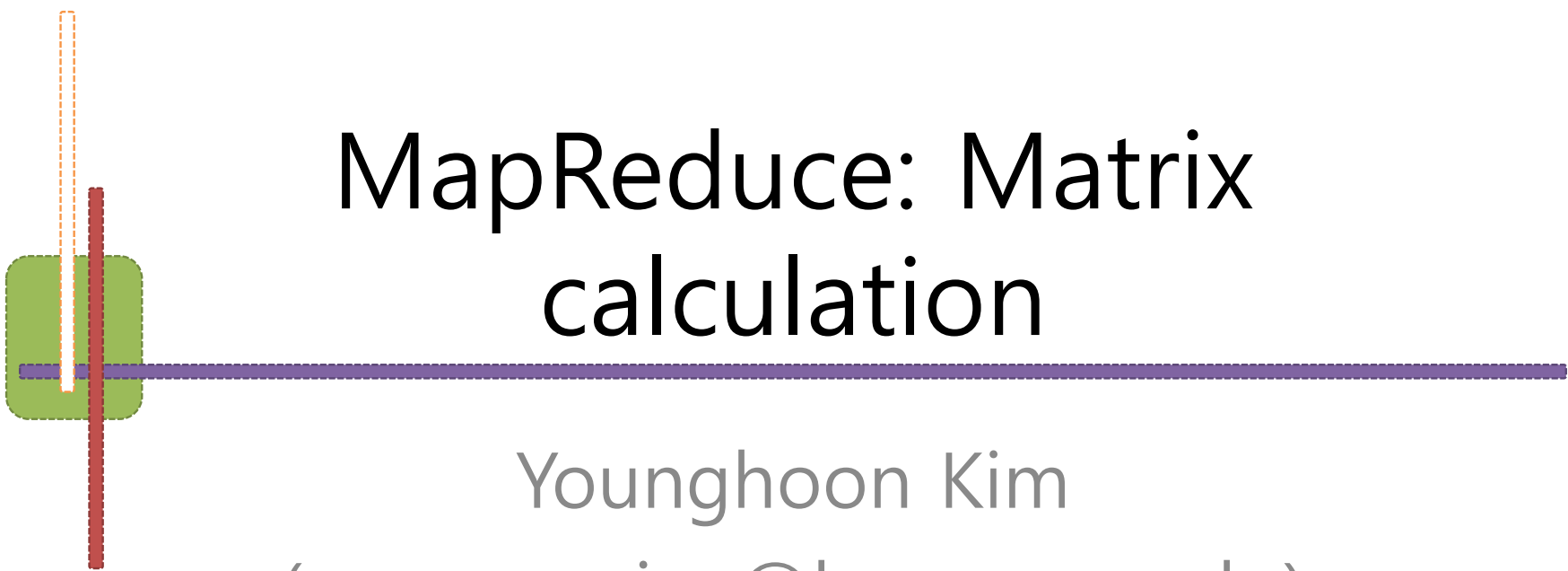




# MapReduce: Matrix calculation

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# **MATRIX ADDITION**



# Matrix Addition: $A + B = C$

$n \times m$     $n \times m$     $n \times m$

## ■ Input

- Two matrices, A and B
- Formatted: a line has a single element
- E.g.,
  - $A, 3, 2, 4.3 \rightarrow A_{32} = 4.3$

## ■ Map

- Input:  $\langle [A|B], i, j, \text{value} \rangle$
- Output:  $\langle \text{key} = \underline{\hspace{2cm}}, \text{value} = \underline{\hspace{2cm}} \rangle$

(1)

(3)

## ■ Reduce

- Input:  $\langle \text{key}, \text{a list of values} \rangle$
- Output:  $\langle \text{key} = \underline{\hspace{2cm}}, \text{value} = \underline{\hspace{2cm}} \rangle$

(2)

(4)



# Matrix Addition: $A + B = C$

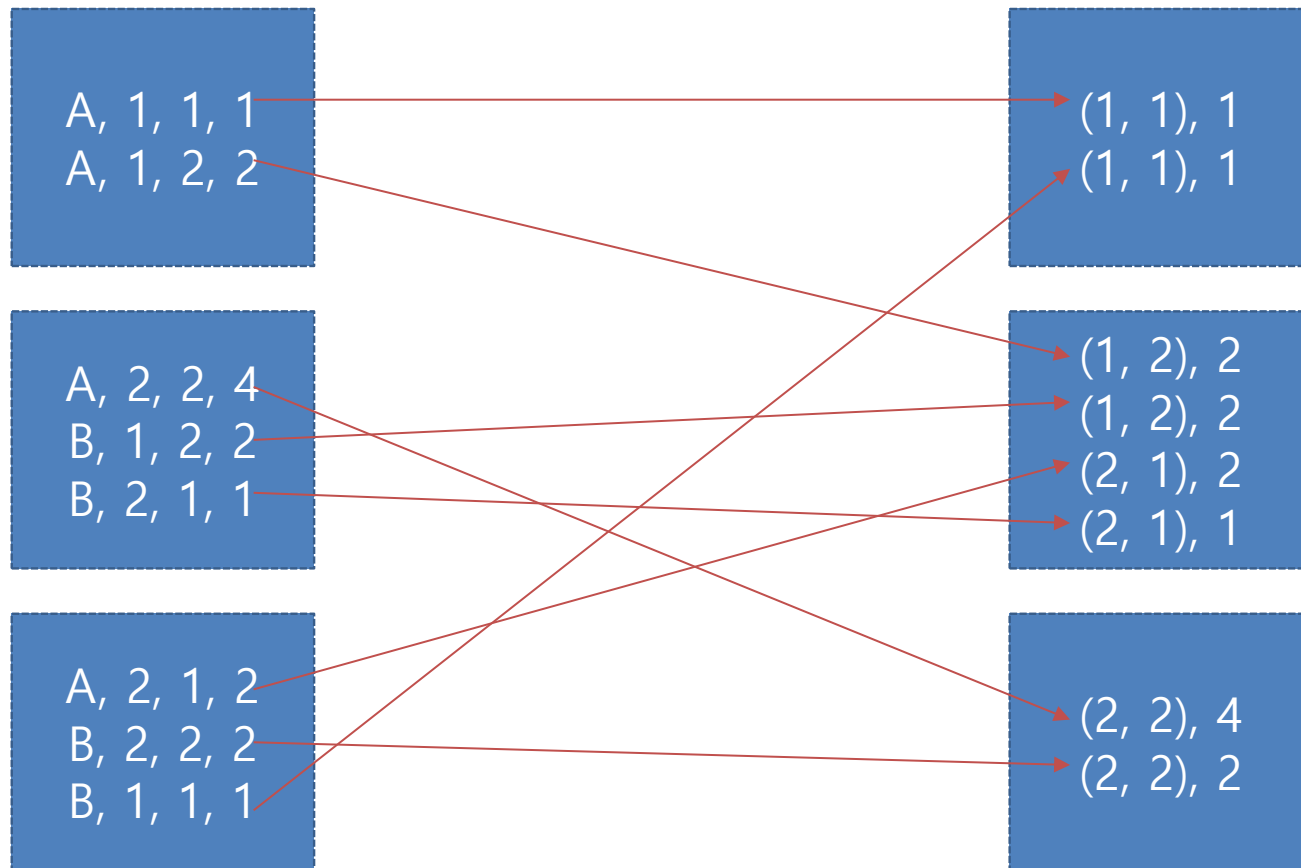
---

- Input
  - Two matrices, A and B
- Map
  - Input:  $\langle [A|B], i, j, \text{value} \rangle$
  - Output:  $\langle \text{key}=\{i,j\}, \text{value}=[A_{ij}|B_{ij}] \rangle$
- Reduce
  - Input:  $\langle \text{key}, \text{a list of values}=\{A_{ij}, B_{ij}\} \rangle$
  - Output:  $\langle \text{key}=\{i,j\}, \text{value}=A_{ij}+B_{ij} \rangle$

(i, j)  
of C

# Matrix Addition: $A + B = C$

```
RDD.map(lambda t: ((t[1], t[2]), t[3]))
```



```
.reduceByKey(lambda a,b: a+b)
```



# Matrix Addition in PySpark

```
!pip install pyspark
```

```
Collecting pyspark  
  Downloading pyspark-3.5.3.tar.gz (317.3 MB)
```

317.3/317.3 MB 3.9

```
  Preparing metadata (setup.py) ... done  
Requirement already satisfied: py4j==0.10.9.7 in /usr/local/lib/python3.10/dist-packages (from pyspark) (  
Building wheels for collected packages: pyspark  
  Building wheel for pyspark (setup.py) ... done  
  Created wheel for pyspark: filename=pyspark-3.5.3-py2.py3-none-any.whl size=317840625 sha256=27e7ffdae6  
  Stored in directory: /root/.cache/pip/wheels/1b/3a/92/28b93e2fbfdbb07509ca4d6f50c5e407f48dce4ddbda69a4a  
Successfully built pyspark  
Installing collected packages: pyspark  
Successfully installed pyspark-3.5.3
```

```
[3] from pyspark.sql import SparkSession
```

```
spark = SparkSession.builder  
    .master("local[*]")  
    .getOrCreate()  
sc = spark.sparkContext
```



# Matrix Addition in PySpark

```
[35] !pip install wget
```



Collecting wget

Downloading wget-3.2.zip (10 kB)

Preparing metadata (setup.py) ... done

Building wheels for collected packages: wget

Building wheel for wget (setup.py) ... done

Created wheel for wget: filename=wget-3.2-py3-none-any.whl size=9656 sha256=d3f428b2836a938f280a3924dff4eb04e9dbd6489aebb5d13f9

Stored in directory: /root/.cache/pip/wheels/8b/f1/7f/5c94f0a7a505ca1c81cd1d9208ae2064675d97582078e6c769

Successfully built wget

Installing collected packages: wget

Successfully installed

"https://raw.githubusercontent.com/nongaussian/class-2024-datamining/refs/heads/main/data/matrix/A200x100.txt"



```
import wget
```

```
url = "https://raw.githubusercontent.com/nongaussian/class-2024-datamining/refs/heads/main/data/matrix/A200x100.txt"
```

```
local_path = 'A200x100.txt'
```

```
wget.download(url, local_path)
```

```
url = "https://raw.githubusercontent.com/nongaussian/class-2024-datamining/refs/heads/main/data/matrix/B200x100.txt"
```

```
local_path = 'B200x100.txt'
```

```
wget.download(url, local_path)
```

"https://raw.githubusercontent.com/nongaussian/class-2024-datamining/refs/heads/main/data/matrix/B200x100.txt"

```
A = sc.textFile("A200x100.txt")
```

```
B = sc.textFile("B200x100.txt")
```



# Matrix Addition in PySpark

```
rdda = A.map(lambda line: line.split(','))\
        .map(lambda t: ((int(t[1]), int(t[2])), float(t[3])))
```

```
rddb = B.map(lambda line: line.split(','))\
        .map(lambda t: ((int(t[1]), int(t[2])), float(t[3])))
```

1) Merging by using union & print out the first tuple

```
rdda.union(rddb).reduceByKey(lambda a, b: a + b)\
    .map(lambda x: ('C', x[0][0], x[0][1], x[1])).first()
```

2) Merging by using join & print out the first tuple

```
rdda.join(rddb)\
    .map(lambda x: ('C', x[0][0], x[0][1], x[1][0] + x[1][1]))\
    .first()
```





# Thinking in MapReduce

---

1. Define the input for mapper function
  - "A, 3, 2, 4.3"
2. Define the key for reducer function
  - Index (i, j) of output matrix C
3. Define the key-value pair to be emitted by mapper (= define mapper function)
  - ((3, 2), 4.3)
4. Define reducer function
  - Sum of list

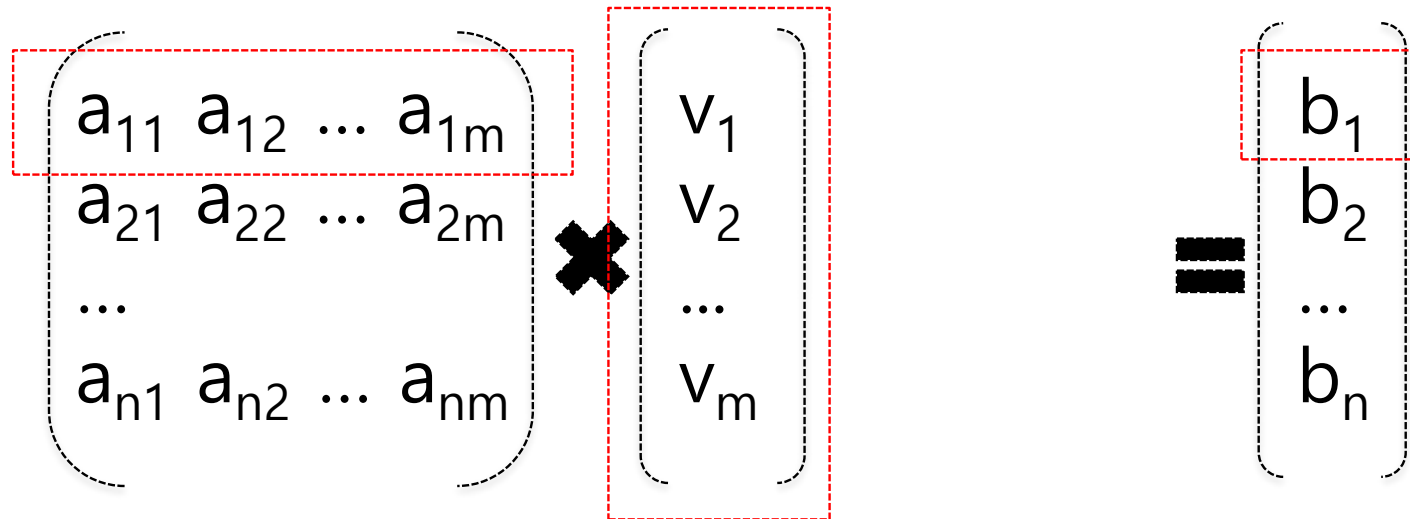
# **MATRIX-VECTOR MULTIPLICATION**



# Matrix Multiplication: $Av=b$

---

- Multiply a matrix and a vector
  - $A * v = b$


$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \times \begin{pmatrix} v_1 \\ v_2 \\ \dots \\ v_m \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{pmatrix}$$



# MapReduce Algorithm

---

## ■ Input

- A n-by-m matrices  $A$  ( $\rightarrow A, [\text{row \#}], [\text{col \#}], [\text{val}]$ )
- A m-dimensional vector  $v$  ( $\rightarrow v, [\text{row \#}], [\text{val}]$ )
- E.g.,
  - $A, 3, 2, 4.3 \rightarrow a_{32} = 4.3$
  - $v, 10, 3.7 \rightarrow v_{10} = 3.7$

## ■ Map

- Input:  $\langle [A|v], i, [j]?, \text{value} \rangle$
- Output:  $\langle \text{key} = ?, \text{value} = ? \rangle$

(1)

## ■ Reduce

- Input:  $\langle \text{key}, \text{a list of values} = \{?\} \rangle$
- Output:  $\langle \text{key} = \text{the row number } i, \text{value} = (Av)_i \rangle$

(2)



# Thinking in MapReduce

---

1. Define the input for mapper function
  - "A, 3, 2, 4.3" or "v, 1, 1.1"
2. Define the key for reducer function
  - The row number  $i$  of output vector  $b$
3. Define the key-value pair to be emitted by mapper
  - $(i, \text{all data to compute } b[i] \text{ if it has})$
4. Define reducer function
  - Compute  $b[i]$

# Matrix Multiplication

- Multiply a matrix and a vector

—  $A * v$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \times \begin{pmatrix} v_1 \\ v_2 \\ \dots \\ v_m \end{pmatrix} =$$

$\langle 1, (A, 1, a_{11}) \rangle$

# Matrix Multiplication

- Multiply a matrix and a vector

—  $A * v$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \times \begin{pmatrix} v_1 \\ v_2 \\ \dots \\ v_m \end{pmatrix} =$$

$\langle 1, (A, 1, a_{11}) \rangle$   
 $\langle 1, (A, 2, a_{12}) \rangle$

# Matrix Multiplication

- Multiply a matrix and a vector

—  $A * v$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \times \begin{pmatrix} v_1 \\ v_2 \\ \dots \\ v_m \end{pmatrix} = \begin{pmatrix} \langle 1, (A, 1, a_{11}) \rangle \\ \langle 1, (A, 2, a_{12}) \rangle \\ \dots \\ \langle 1, (A, m, a_{1m}) \rangle \end{pmatrix}$$



# Matrix Multiplication

- Multiply a matrix and a vector

—  $A * v$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \times \begin{pmatrix} v_1 \\ v_2 \\ \dots \\ v_m \end{pmatrix} = \begin{pmatrix} \langle 1, (A, 1, a_{11}) \rangle \\ \langle 1, (A, 2, a_{12}) \rangle \\ \dots \\ \langle 1, (A, m, a_{1m}) \rangle \\ \langle 1, (v, 1, v_1) \rangle \\ \dots \\ \langle 2, (v, 1, v_1) \rangle \\ \dots \\ \langle n, (v, 1, v_1) \rangle \end{pmatrix}$$

A red dashed arrow points from the  $v_1$  element in the vector to the first row of the resulting vector.

# Matrix Multiplication

- Multiply a matrix and a vector

—  $A * v$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \times \begin{pmatrix} v_1 \\ v_2 \\ \dots \\ v_m \end{pmatrix} = \begin{pmatrix} \langle 1, (A, 1, a_{11}) \rangle \\ \langle 1, (A, 2, a_{12}) \rangle \\ \dots \\ \langle 1, (A, m, a_{1m}) \rangle \\ \langle 1, (v, 1, v_1) \rangle \\ \langle 1, (v, 2, v_2) \rangle \\ \dots \\ \langle n, (v, 1, v_1) \rangle \\ \langle n, (v, 2, v_2) \rangle \end{pmatrix}$$

The diagram illustrates the multiplication of a matrix  $A$  (with elements  $a_{ij}$ ) and a vector  $v$  (with elements  $v_i$ ). The result is a vector where each element is a dot product of a row of  $A$  with the vector  $v$ . A red dashed circle highlights the element  $v_1$  in the vector  $v$ , and a red dashed arrow points from it to the first row of the resulting vector, indicating the calculation of the first row's dot product.

# Matrix Multiplication

- Multiply a matrix and a vector

—  $A * v$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \times \begin{pmatrix} v_1 \\ v_2 \\ \dots \\ v_m \end{pmatrix} = \begin{pmatrix} \langle 1, (A, 1, a_{11}) \rangle \\ \langle 1, (A, 2, a_{12}) \rangle \\ \dots \\ \langle 1, (A, m, a_{1m}) \rangle \\ \langle 1, (v, 1, v_1) \rangle \\ \langle 1, (v, 2, v_2) \rangle \\ \dots \\ \langle 1, (v, m, v_m) \rangle \\ \langle 2, (v, 1, v_1) \rangle \\ \langle 2, (v, 2, v_2) \rangle \\ \dots \\ \langle 2, (v, m, v_m) \rangle \\ \dots \\ \langle n, (v, 1, v_1) \rangle \\ \langle n, (v, 2, v_2) \rangle \\ \dots \\ \langle n, (v, m, v_m) \rangle \end{pmatrix}$$

The diagram illustrates the multiplication of a matrix  $A$  (with elements  $a_{ij}$ ) and a vector  $v$  (with elements  $v_i$ ). The result is a vector of inner products. A red dashed circle highlights the element  $v_m$  in the vector, and a red dashed arrow points from it to the corresponding element in the resulting vector, indicating the dot product of the first row of the matrix with the vector  $v$ .

# Matrix Multiplication

- Multiply a matrix and a vector

—  $A * v$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix} \times \begin{pmatrix} v_1 \\ v_2 \\ \dots \\ v_m \end{pmatrix} = \begin{pmatrix} \langle 1, (A, 1, a_{11}) \rangle \\ \langle 1, (A, 2, a_{12}) \rangle \\ \dots \\ \langle 1, (A, m, a_{1m}) \rangle \\ \langle 1, (v, 1, v_1) \rangle \\ \langle 1, (v, 2, v_2) \rangle \\ \dots \\ \langle 1, (v, m, v_m) \rangle \\ \langle 2, (v, 1, v_1) \rangle \\ \langle 2, (v, 2, v_2) \rangle \\ \dots \\ \langle 2, (v, m, v_m) \rangle \\ \dots \\ \langle n, (v, 1, v_1) \rangle \\ \langle n, (v, 2, v_2) \rangle \\ \dots \\ \langle n, (v, m, v_m) \rangle \end{pmatrix}$$



# MapReduce Algorithm

---

## ■ Map

- Input:  $\langle [A|v], i, [j]?, [a_{ij} \mid v_i] \rangle$
- Output:
  - If "A": Output  $\langle \text{key}=i, \text{value}= ("A", j, a_{ij}) \rangle$
  - Else if "v":
    - ◆ For  $k=1$  to  $n$ :
      - » Output  $\langle \text{key}=k, \text{value}= ("v", i, v_i) \rangle$

## ■ Reduce

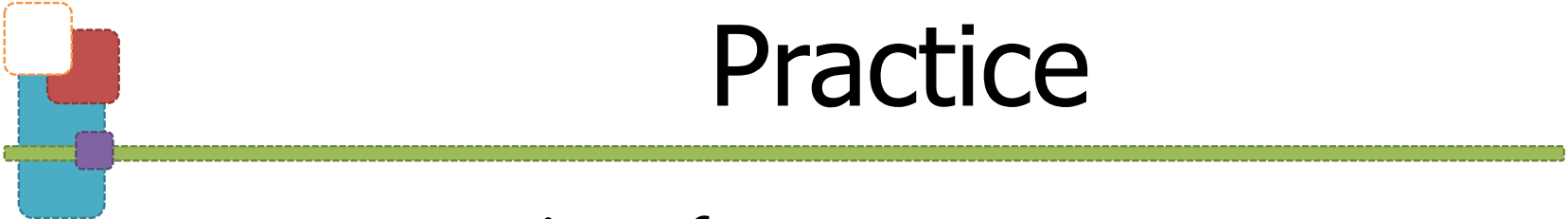
- Input:  $\langle \text{key}=i,$   
a list of values  $= \{(A, j, a_{ij}) \mid (v, j, v_j)\}_{j=1, \dots, m} \rangle$
- Output:  $\langle \text{key}=i, \text{value} = \sum_{j=1}^m a_{ij} \cdot v_j \rangle$



# Practice

- Download

- `https://raw.githubusercontent.com/nongaussian/class-2024-datamining/refs/heads/main/data/matrix/A200x100.txt`
- `https://raw.githubusercontent.com/nongaussian/class-2024-datamining/refs/heads/main/data/matrix/v100.txt`



# Practice

## ■ Preprocessing for RDDs

```
rdda = A.map(lambda line: line.split(',')\
            .map(
                lambda t: (int(t[1]), ('A', int(t[2]), float(t[3])))))
```

```
rddv = v.map(lambda line: line.split(',')\
            .flatMap(
                lambda t: [(i, ('v', int(t[1]), float(t[2]))) \
                           for i in range(200)])
```

Increasing the keys from 1 to n



# Practice

- Cogroup by the key & compute the inner product

```
b = rdda.cogroup(rddv) \
    .mapValues(
```

Sort the list by index label & compute the inner product using Numpy

```
    lambda l:
        (np.array(sorted(l[0], key=lambda t: t[1]))[:,2] \
         .astype(np.float32)
         * np.array(sorted(l[1], key=lambda t: t[1]))[:,2] \
         .astype(np.float32)))
```

To handle type error...





# Thinking in MapReduce

---

1. Define the input for mapper function
  - "A, 3, 2, 4.3" or "v, 1, 1.1"
2. Define the key for reducer function
  - The row number  $i$  of output vector  $b$
3. Define the key-value pair to be emitted by mapper
  - $(i, \text{all data to compute } b[i] \text{ if it has})$
4. Define reducer function
  - Compute  $b[i]$

# **MATRIX-MATRIX MULTIPLICATION**



# Matrix Multiplication: $A \times B = C$

## Map

### — Input

- Two matrices,  $A$  ( $n \times \ell$ ) and  $B$  ( $\ell \times m$ )

- E.g.,

♦  $A, 2, 3, 3.2 \rightarrow a_{23} = 3.2$  (1)

♦  $B, 2, 3, 3.2 \rightarrow b_{23} = 3.2$

### — Output

- For  $(A, i, j, a_{ij})$ : ?
- For  $(B, i, j, b_{ij})$ : ?

## Reduce

### — Input

- $\langle \text{key}=?, ? \rangle$  (2)

### — Output

- $\langle \text{key}=?, ? \rangle$



# Matrix Multiplication: $A \times B = C$

## Map

### — Input

- Two matrices,  $A$  ( $n \times \ell$ ) and  $B$  ( $\ell \times m$ )

- E.g.,

♦  $A, 2, 3, 3.2 \rightarrow a_{23} = 3.2$  (1)

♦  $B, 2, 3, 3.2 \rightarrow b_{23} = 3.2$

### — Output

- For  $(A, i, j, a_{ij})$ : ? (3)

- For  $(B, i, j, b_{ij})$ : ?

## Reduce

### — Input

- $\langle \text{key}=\{i,j\}, \{?\} \rangle$  (2)

### — Output

- $\langle \text{key}=\{i,j\}, ? \rangle$



# Matrix Multiplication

---

- A matrix: row 1 -> C matrix: row 1

Decide to which reducers  
we send the data

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



# Matrix Multiplication

---

- A matrix: row 1  $\rightarrow$  C matrix: row 1

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



# Matrix Multiplication

---

- A matrix: row 1 -> C matrix: row 1

Each element should be sent to the reducer of  $C_{11}, C_{12}, \dots, C_{1m}$ .

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



# Matrix Multiplication

---

- A matrix: row 2  $\rightarrow$  C matrix: row 2

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$





# Matrix Multiplication

- A matrix: row 2  $\rightarrow$  C matrix: row 2

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



# Matrix Multiplication

---

- A matrix: row 2 -> C matrix: row 2

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



# Matrix Multiplication

- A matrix i-th row  $\rightarrow$  C matrix i-th row

$a_{ij} \rightarrow ((i, 1), (j, a_{ij})), ((i, 2), (j, a_{ij})), \dots, ((i, m), (j, a_{ij}))$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



# Matrix Multiplication

---

- B matrix: col 1  $\rightarrow$  C matrix: col 1

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



# Matrix Multiplication

---

- B matrix: col 1  $\rightarrow$  C matrix: col 1

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



# Matrix Multiplication

---

- B matrix: col 1  $\rightarrow$  C matrix: col 1

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



# Matrix Multiplication

---

- B matrix: col 2 -> C matrix: col 2

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



# Matrix Multiplication

- B matrix: col 2 -> C matrix: col 2

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$





# Matrix Multiplication

---

- B matrix: col 2 -> C matrix: col 2

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



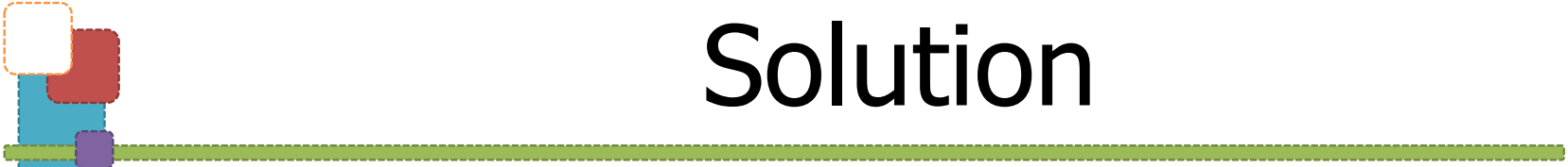
# Matrix Multiplication

---

- B matrix j-th col  $\rightarrow$  C matrix j-th col

$b_{ij} \rightarrow ((1, j), (i, b_{ij})), ((2, j), (i, b_{ij})), \dots, ((n, j), (i, b_{ij}))$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nk} \end{pmatrix} \times \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & & & \\ b_{k1} & b_{k2} & \dots & b_{km} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & & & \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix}$$



# Solution

## ■ Map

### — Input

- Two matrices,  $A$  ( $n \times l$ ) and  $B$  ( $l \times m$ )

### — Output

- For  $A_{ij}$ :  $\langle \{i, 1\}, \{j, A_{ij}\} \rangle, \dots, \langle \{i, m\}, \{j, A_{ij}\} \rangle$
- For  $B_{ij}$ :  $\langle \{1, j\}, \{i, B_{ij}\} \rangle, \dots, \langle \{n, j\}, \{i, B_{ij}\} \rangle$

## ■ Reduce

### — Input

- $\langle \{i, j\}, \{?\} \rangle$

### — Output

- $\langle \{i, j\}, ? \rangle$



# Solution

## ■ Map

### — Input

- Two matrices,  $A$  ( $n \times l$ ) and  $B$  ( $l \times m$ )

### — Output

- For  $A_{ij}$ :  $\langle \{i, 1\}, \{j, A_{ij}\} \rangle, \dots, \langle \{i, m\}, \{j, A_{ij}\} \rangle$
- For  $B_{ij}$ :  $\langle \{1, j\}, \{i, B_{ij}\} \rangle, \dots, \langle \{n, j\}, \{i, B_{ij}\} \rangle$

## ■ Reduce

### — Input

- $\langle \{i, j\}, \{\{1, A_{i1}\}, \{1, B_{1j}\}, \dots, \{l, A_{il}\}, \{l, B_{lj}\}\} \rangle$

### — Output

- $\langle \{i, j\}, A_{i1} * B_{1j} + A_{i2} * B_{2j} + \dots + A_{il} * B_{lj} \rangle$

# Matrix Multiplication: Example

- $a_{ik}$ 
  - key  $(i, 1), (i, 2), \dots (i, m)$
  - value  $(k, a_{ik})$

A 1 1 3

key (1,1) value (1,3)

key (1,2) value (1,3)

A 1 2 -5

A 2 1 6

A 2 2 12

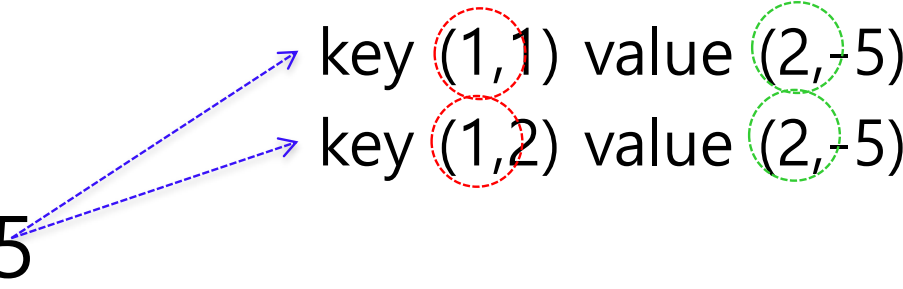
$$\begin{pmatrix} 3 & -5 \\ 6 & 12 \end{pmatrix} \times \begin{pmatrix} 2 & 11 \\ 1 & -7 \end{pmatrix} =$$

# Matrix Multiplication

- $a_{ik}$ 
  - key  $(i, 1), (i, 2), \dots (i, m)$
  - value  $(k, a_{ik})$

A 1 1 3  
A 1 2 -5  
A 2 1 6  
A 2 2 12

key (1,1) value (2,-5)  
key (1,2) value (2,-5)



$$\begin{pmatrix} 3 & -5 \\ 6 & 12 \end{pmatrix} \times \begin{pmatrix} 2 & 11 \\ 1 & -7 \end{pmatrix} =$$

# Matrix Multiplication

- $a_{ik}$ 
  - key  $(i, 1), (i, 2), \dots (i, m)$
  - value  $(k, a_{ik})$

A 1 1 3

A 1 2 -5

A 2 1 6

A 2 2 12

key (2,1) value (1,6)

key (2,2) value (1,6)

$$\begin{pmatrix} 3 & -5 \\ 6 & 12 \end{pmatrix} \times \begin{pmatrix} 2 & 11 \\ 1 & -7 \end{pmatrix} =$$

# Matrix Multiplication

- $a_{ik}$ 
  - key  $(i, 1), (i, 2), \dots (i, m)$
  - value  $(k, a_{ik})$

A 1 1 3

A 1 2 -5

A 2 1 6

A 2 2 12

key (2,1) value (2,12)

key (2,2) value (2,12)

$$\begin{pmatrix} 3 & -5 \\ 6 & 12 \end{pmatrix} \times \begin{pmatrix} 2 & 11 \\ 1 & -7 \end{pmatrix} =$$



# Matrix Multiplication

- $b_{kj}$ 
  - key  $(1, j), (2, j), \dots (n, j)$
  - value  $(k, b_{kj})$

B  $\begin{matrix} \textcircled{1} & \textcircled{1} & 2 \end{matrix}$   $\begin{matrix} \nearrow \text{key } (1, \textcircled{1}) \text{ value } \textcircled{(1,2)} \\ \searrow \text{key } (2, \textcircled{1}) \text{ value } \textcircled{(1,2)} \end{matrix}$

B 1 2 11

B 2 1 1

B 2 2 -7

$$\begin{pmatrix} 3 & -5 \\ 6 & 12 \end{pmatrix} \times \begin{pmatrix} 2 & 11 \\ 1 & -7 \end{pmatrix} =$$

# Matrix Multiplication

- $b_{kj}$ 
  - key  $(1, j), (2, j), \dots (n, j)$
  - value  $(k, b_{kj})$

B 1 1 2  
B 1 2 1 1  
B 2 1 1  
B 2 2 -7

key (1,2) value (1,11)  
key (2,2) value (1,11)

$$\begin{pmatrix} 3 & -5 \\ 6 & 12 \end{pmatrix} \times \begin{pmatrix} 2 & 11 \\ 1 & -7 \end{pmatrix} =$$

# Matrix Multiplication

- $b_{kj}$ 
  - key  $(1, j), (2, j), \dots (n, j)$
  - value  $(k, b_{kj})$

B 1 1 2

B 1 2 11

B 2 1 1

B 2 2 -7

key (1, 1) value (2, 1)

key (2, 1) value (2, 1)

$$\begin{pmatrix} 3 & -5 \\ 6 & 12 \end{pmatrix} \times \begin{pmatrix} 2 & 11 \\ 1 & -7 \end{pmatrix} =$$

# Matrix Multiplication

- $b_{kj}$ 
  - key  $(1, j), (2, j), \dots (n, j)$
  - value  $(k, b_{kj})$

B 1 1 2

B 1 2 11

B 2 1 1

B 2 2 -7

key (1, 2) value (2, -7)

key (2, 2) value (2, -7)

$$\begin{pmatrix} 3 & -5 \\ 6 & 12 \end{pmatrix} \times \begin{pmatrix} 2 & 11 \\ 1 & -7 \end{pmatrix} =$$



# Matrix Multiplication

A emitted  
key value

|       |        |
|-------|--------|
| (1,1) | (1,3)  |
| (1,2) | (1,3)  |
| (1,1) | (2,-5) |
| (1,2) | (2,-5) |
| (2,1) | (1,6)  |
| (2,2) | (1,6)  |
| (2,1) | (2,12) |
| (2,2) | (2,12) |

B emitted  
key value

|       |        |
|-------|--------|
| (1,1) | (1,2)  |
| (2,1) | (1,2)  |
| (1,2) | (1,11) |
| (2,2) | (1,11) |
| (1,1) | (2,1)  |
| (2,1) | (2,1)  |
| (1,2) | (2,-7) |
| (2,2) | (2,-7) |

$$\begin{pmatrix} 3 & -5 \\ 6 & 12 \end{pmatrix} \times \begin{pmatrix} 2 & 11 \\ 1 & -7 \end{pmatrix} =$$

# Matrix Multiplication

A emitted  
key value

(1,1) (1,3)  
(1,2) (1,3)  
(1,1) (2,-5)  
(1,2) (2,-5)  
(2,1) (1,6)  
(2,2) (1,6)  
(2,1) (2,12)  
(2,2) (2,12)

B emitted  
key value

(1,1) (1,2)  
(2,1) (1,2)  
(1,2) (1,11)  
(2,2) (1,11)  
(1,1) (2,1)  
(2,1) (2,1)  
(1,2) (2,-7)  
(2,2) (2,-7)

key

(1,1)

value list  
<(1,3) , (2,-5), (1, 2), (2, 1)>

(1,2)

<(1,3), (2,-5), (1, 11), 2,-7)>

(2,1)

<(1,6) , (2,12), (1, 2), (2, 1)>

(2,2)

<(1,6) , (2,12), (1, 11), (2, -7)>

$$\begin{pmatrix} 3 & -5 \\ 6 & 12 \end{pmatrix} \times \begin{pmatrix} 2 & 11 \\ 1 & -7 \end{pmatrix} =$$

# Matrix Multiplication

A emitted  
key value

(1,1) (1,3)  
(1,2) (1,3)  
(1,1) (2,-5)  
(1,2) (2,-5)  
(2,1) (1,6)  
(2,2) (1,6)  
(2,1) (2,12)  
(2,2) (2,12)

B emitted  
key value

(1,1) (1,2)  
(2,1) (1,2)  
(1,2) (1,11)  
(2,2) (1,11)  
(1,1) (2,1)  
(2,1) (2,1)  
(1,2) (2,-7)  
(2,2) (2,-7)

key

value list

(1,1) <(1,3), (2,-5), (1,2), (2,1)>

$$6 - 5 = 1$$

(1,2) <(1,3), (2,-5), (1,11), (2,-7)>

$$33 + 35 = 68$$

(2,1) <(1,6), (2,12), (1,2), (2,1)>

$$12 + 12 = 24$$

(2,2) <(1,6), (2,12), (1,11), (2,-7)>

$$66 - 84 = -18$$

$$\begin{pmatrix} 3 & -5 \\ 6 & 12 \end{pmatrix} \times \begin{pmatrix} 2 & 11 \\ 1 & -7 \end{pmatrix} = \begin{pmatrix} 1 & 68 \\ 24 & -18 \end{pmatrix}$$



# Practice

---

- Download

- `https://raw.githubusercontent.com/nongaussian/class-2024-datamining/refs/heads/main/data/matrix/A200x100.txt`
- `https://raw.githubusercontent.com/nongaussian/class-2024-datamining/refs/heads/main/data/matrix/B100x10.txt`





# Thinking in MapReduce

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1. Define the input for mapper function
2. Define the key for reducer function
3. Define the key-value pair to be emitted by mapper
4. Define reducer function

# DISCUSSION



# Alternative Solution (1)

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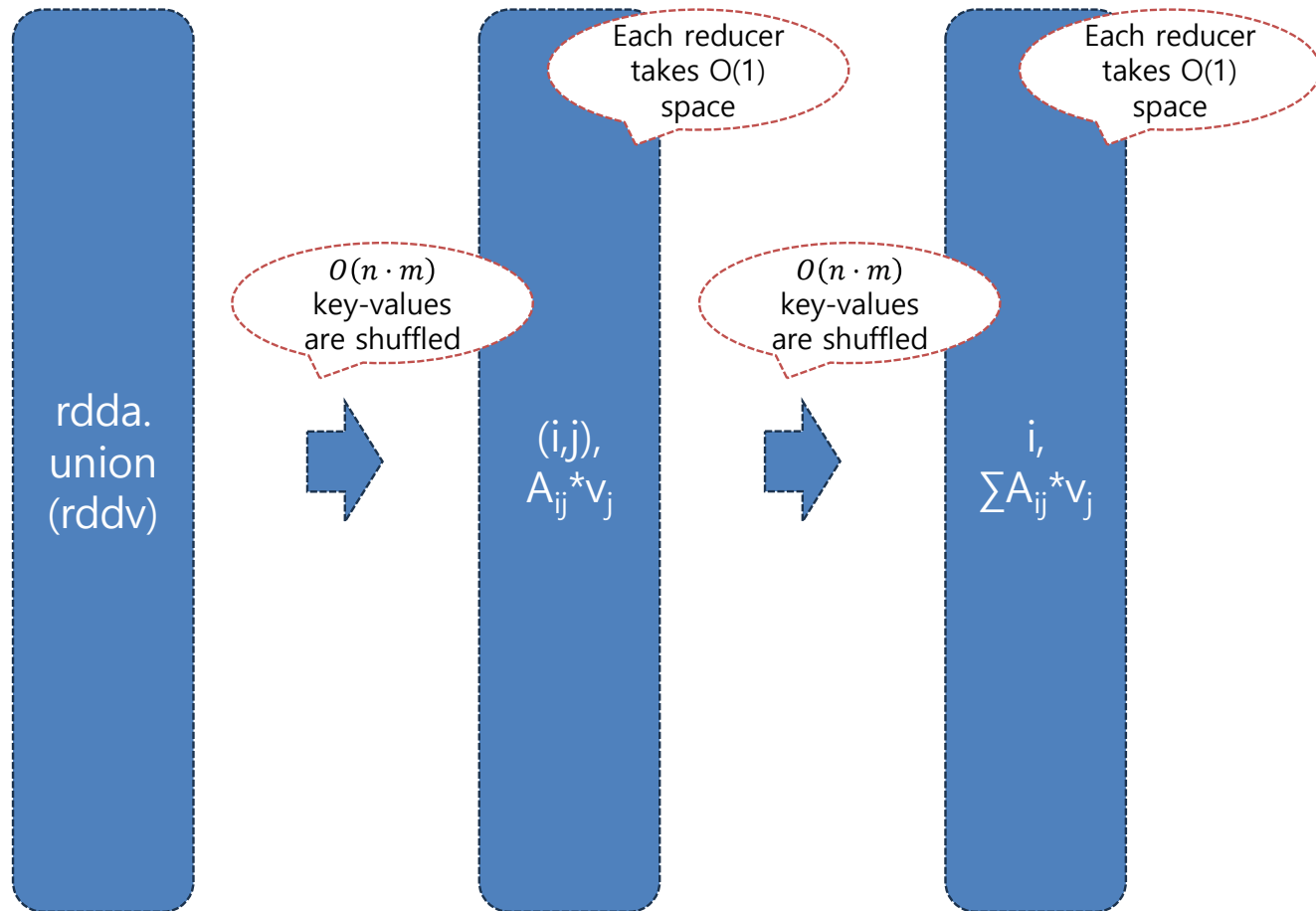
- Consider the following excellent code for mat-vec multiplication submitted by a student

```
rdda = A.map(lambda line: line.split(',')\
        .map(lambda t : ((int(t[1]), int(t[2])), float(t[3]))))
rddv = v.map(lambda line: line.split(',')\
        .flatMap(lambda t :
        [((k, int(t[1])), float(t[2])) for k in range(0, v_count)]))

somav = rdda.union(rddv)\
        .reduceByKey(lambda a, b: a * b)\
        .map(lambda t : (t[0][0], t[1]))\
        .reduceByKey(lambda x, y : x+y)
```

# Alternative Solution (2)

- Count the shuffling





# Alternative Solution (2)

---

- The solution using cogroup we examined in our course

```
b = rdda.cogroup(rddv).mapValues(  
    lambda l: (  
        np.dot(np.array(sorted(l[0], key=lambda t: t[1]))[:,2]\  
            .astype(np.float32),  
            np.array(sorted(l[1], key=lambda t: t[1]))[:,2]\  
            .astype(np.float32))  
    ))
```

# Alternative Solution (2)

- Count the shuffling

